

CAUSALITY ASSESSMENT

Pharmacovigilance Internship Task — VirtualWorks

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Standard Causality Assessment Methods

Method	Used By	Output Categories
Naranjo Algorithm	Clinical researchers, CROs	Definite / Probable / Possible / Doubtful (Score-based: ≥ 9 / 5-8 / 1-4 / ≤ 0)
WHO-UMC Scale	Regulatory agencies, CDSCO	Certain / Probable / Possible / Unlikely / Unclassified
RUCAM Scale	Hepatotoxicity assessment	Highly probable / Probable / Possible / Unlikely / Excluded

CASE 1 — Amoxicillin-Clavulanate & Maculopapular Skin Rash

Case Scenario

45-year-old female, 62 kg. Prescribed Augmentin 625 mg BD for community-acquired RTI. On Day 3, developed diffuse erythematous maculopapular rash (trunk + upper limbs), mild facial edema, pruritus. Drug discontinued immediately. Treated with Cetirizine 10 mg oral + topical corticosteroid. Rash resolved completely in 5 days. No hospitalization. Known childhood sulfonamide hypersensitivity.

Naranjo Algorithm

#	Question	Score
1	Previous conclusive reports of this reaction?	+1
2	ADR appeared after the suspected drug was administered?	+2
3	ADR improved when drug was discontinued?	+1
4	ADR reappeared on rechallenge? (Not performed — clinically inappropriate)	0
5	Alternative causes ruled out?	+1
6	ADR confirmed by placebo? (N/A)	0
7	Drug detected in toxic concentration in blood/fluids?	0
8	Dose-response relationship demonstrated?	0
9	Prior similar reaction to same/related drug? (Sulfonamide history — cross-reactivity risk)	+1
10	Confirmed by objective evidence? (Physician-confirmed rash, clinical exam)	+1
TOTAL		+7

WHO-UMC Scale

WHO-UMC Criteria	Met?
Plausible time relationship (Day 3 — known latency for beta-lactam hypersensitivity)	✓ Yes
Known reaction type for the drug (listed in Augmentin SmPC)	✓ Yes
Reaction improved on dechallenge (resolved in 5 days)	✓ Yes
No plausible alternative cause (no other drug, no infection-related rash)	✓ Yes
Rechallenge confirmed (not performed — ethically inappropriate)	✗ No

WHO-UMC Classification: PROBABLE — Certain requires rechallenge confirmation

Key Learning: Rechallenge is ethically inappropriate in allergic reactions — 'Probable' is the highest achievable classification in spontaneous reports without rechallenge.

CASE 2 — Metformin & Lactic Acidosis

Case Scenario

58-year-old male, Type 2 Diabetes. On Metformin 1000 mg BD x 6 months. Admitted with confusion, rapid breathing, severe fatigue. Blood lactate: 8.2 mmol/L (normal <2 mmol/L). Serum creatinine elevated — eGFR 28 (Stage 3b CKD, previously undetected). Metformin stopped; IV sodium bicarbonate given. Recovered in 4 days. No other suspect drug.

Naranjo Algorithm

#	Question	Score
1	Previous conclusive reports? (Metformin-associated lactic acidosis is well-documented)	+1
2	ADR after drug administration? (Risk accumulated with declining renal function)	+2

3	ADR improved on discontinuation? (Recovered after stopping + treatment)	+1
4	Rechallenge performed? (Contraindicated — CKD Stage 3b present)	0
5	Alternative cause ruled out? (No sepsis, no other cause of lactic acidosis found)	+1
6	Placebo reaction? (N/A)	0
7	Drug detected in toxic concentration? (Renal impairment → Metformin accumulation likely)	+1
8	Dose-response relationship demonstrated?	0
9	Prior similar reaction to same drug?	0
10	Confirmed by objective evidence? (Lactate 8.2 mmol/L — biochemically confirmed)	+1
	TOTAL	

WHO-UMC Scale

WHO-UMC Criteria	Met?
Plausible time sequence (chronic accumulation with renal function decline)	✓ Yes
Known reaction — Metformin contraindicated in eGFR <30	✓ Yes
Reaction improved on dechallenge	✓ Yes
No plausible alternative cause	✓ Yes
Rechallenge performed?	✗ No
WHO-UMC Classification: PROBABLE	

Key Learning: Metformin lactic acidosis is a class effect, not idiosyncratic. The real PV signal here is prescribing in undetected CKD — a medication error + ADR overlap scenario that warrants prescriber education.

CASE 3 — Atorvastatin & Myalgia (Ambiguous Case)

Case Scenario

52-year-old male. Hypertension + Hyperlipidemia. On Atorvastatin 40 mg OD x 3 weeks + Amlodipine 5 mg OD. Reports bilateral leg pain and muscle weakness. CK: 280 U/L (ULN: 200 U/L — mildly elevated; not rhabdomyolysis threshold). Patient also started a new gym routine 10 days ago. Dose reduced to Atorvastatin 20 mg — pain partially improved. Drug not stopped completely.

Naranjo Algorithm

#	Question	Score
1	Previous conclusive reports? (Statin myopathy is well-documented)	+1
2	ADR after drug administration?	+2
3	Improved on discontinuation/dose reduction? (Partial improvement on dose reduction)	+1
4	Rechallenge performed?	0
5	Alternative causes ruled out? (NO — gym exercise is a plausible confounding cause)	-1
6	Placebo reaction? (N/A)	0
7	Toxic drug concentration? (CK only mildly elevated — not diagnostic)	0
8	Dose-response relationship? (40mg → 20mg → partial improvement suggests dose-response)	+1
9	Prior similar reaction to same drug?	0
10	Confirmed by objective evidence? (CK mildly elevated but not diagnostic alone)	0
	TOTAL	

WHO-UMC Scale

WHO-UMC Criteria	Met?
Plausible time relationship (3 weeks — consistent with statin myopathy onset)	✓ Yes
Known reaction (listed in Atorvastatin SmPC)	✓ Yes
Improved on dechallenge (partial only — not stopped completely)	■ Partial
No plausible alternative cause (FAILED — exercise is a significant confounder)	✗ No
Rechallenge performed?	✗ No
WHO-UMC Classification: POSSIBLE — Confounding factor (exercise) prevents upgrade to Probable	

Key Learning: Exercise + Statin = classic confounding scenario. To upgrade causality, a 2-week drug holiday + CK retest + rechallenge is required. Always document and rule out lifestyle confounders in musculoskeletal ADRs.

Summary — All 3 Cases					
Case	Drug	Reaction	Naranjo Score	Naranjo Class	WHO-UMC
1	Amoxicillin-Clavulanate (Augmentin 625 mg)	Maculopapular rash, facial edema	+7	PROBABLE	PROBABLE
2	Metformin 1000 mg BD	Lactic acidosis (lactate 8.2 mmol/L)	+7	PROBABLE	PROBABLE
3	Atorvastatin 40 mg OD	Bilateral myalgia, mild CK elevation	+4	POSSIBLE	POSSIBLE

Quick Reference — Classification Thresholds	
NARANJO SCORE	WHO-UMC SCALE
$\geq 9 \rightarrow$ Definite $5-8 \rightarrow$ Probable $1-4 \rightarrow$ Possible $\leq 0 \rightarrow$ Doubtful	CERTAIN = Rechallenge confirmed + all criteria met PROBABLE = Temporal + known + dechallenge + no alternative cause POSSIBLE = Temporal + known reaction BUT alternative cause present UNLIKELY = Poor temporal relationship + alternative cause likely

Key Exam / Interview Tips	
Concept	Remember
Dechallenge positive	Reaction improves on stopping the drug $\rightarrow$ upgrades causality
Rechallenge positive	Reaction returns on restarting $\rightarrow$ confirms CERTAIN classification
Confounding factor	Alternative cause present $\rightarrow$ always downgrades to POSSIBLE
Naranjo Q5	Most commonly scored wrong — rule out ALL alternative causes carefully
CERTAIN classification	Almost never used in spontaneous reports — requires rechallenge confirmation